

Is this trail too crowded? A choice experiment to evaluate tradeoffs and preferences of park visitors in Garibaldi Park, British Columbia

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Large crowds in parks can be a problem for park managers and visitors. However, perceptions of crowding are difficult to measure due to coping mechanisms deployed by park visitors. Furthermore, perceptions of crowding should not be measured in isolation, but rather as part of a suite of conditions that comprise the visitors' outdoor experience. We used a dichotomous choice experiment with visual images and eight attributes to estimate park users' utilities associated with their visitor experience in Garibaldi Provincial Park in British Columbia, Canada. Our visual method allowed us to control for background view and compare user preferences on hiking trails with preferences at final destinations. **We find that utilities are more sensitive to crowding at viewpoints than to other aspects of the outdoor experience. Thus, visitor satisfaction and crowding perceptions are more likely to be defined by where visitors have these encounters rather than the total number of encounters.**

Keywords: British Columbia; Canada; crowding; leisure; discrete choice experiment; outdoors; park fees; park management; part-worth utility; protected areas; recreation

1. Introduction

Park managers confront a paradoxical mandate: to provide recreational opportunities for the public *while simultaneously* protecting ecosystems and ensuring visitor satisfaction (Manning 2007; Newsome, Moore, and Dowling 2012). This dual objective means that park managers must sometimes make difficult choices between competing goals. Wise management must find a way to balance recreational use with ecosystem conservation, and visitor satisfaction (van Riper *et al.* 2011).

Visitors to parks and recreation areas face a related challenge when selecting outdoor destinations (Cheung and Jim 2013). Many outdoor enthusiasts are motivated to find remote places, distant from human activity (Lawson and Manning 2001). At the same time, recreational users tend to appreciate basic services provided by park managers, such as trail maintenance, temporary shelters, campgrounds, toilets, signs, and interpretive material, even though these useful services also indicate human presence and management (Newman *et al.* 2005). As a result of these conflicting desires, both park managers and visitors alike need to make tradeoffs. Understanding how visitors make these tradeoffs between a desire for solitude versus a desire for public services and amenities can help park managers improve the recreational experience.

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Visitor crowding is one particularly important factor that may determine the satisfaction of a recreational experience (Amberger and Brandenburg 2007; Manning *et al.* 1996; Manning *et al.* 2002; Pettebone *et al.* 2013; Vaske and Shelby 2008). Large crowds in a park or protected area can negatively impact visitor experience, but limiting crowds might contradict the mandate of managers, who are expected to ensure that parks are widely accessible to the public. Therefore, the management choice to limit visitors in order to reduce the negative effects of crowding (e.g. through the use of user fees) might be contentious (Manning 2003; Stewart and Cole 2001; Wagar 1964). In recent decades, visitor crowding has become a widely debated subject in recreational management, and has even been called one of the most “fundamental and intensely studied issues in outdoor recreation” (Manning and Valliere 2001, 410).

In this paper, we aim to make several contributions to the literature on the planning and management of parks and protected areas. In particular, we aim to understand the tradeoffs that visitors are willing to make to achieve a satisfactory outdoor experience. Our research is unique in that we use a dichotomous choice experiment and latent class analysis to understand the heterogeneity of visitor preferences and tradeoffs for different user types. Latent class analysis has been used in studies on recreational fisheries (Beardmore, Haider, Hunt, and Arlinghaus 2013), recreational hunting (Hunt, Haider, and Botton 2005), cross country ski resorts (Landauer, Pröbstl, and Haider 2012), and urban parks (Amberger *et al.* 2010), but surprisingly few studies have employed latent class analysis to understand the heterogeneity of protected area visitor preferences, with only a few exceptions (Juutinen *et al.* 2011; Wallmo and Edwards 2008). From a management perspective, understanding visitor preferences may help guide decisions about park access, user fees, crowd management, or other relevant policies.

This paper also adds to the literature on perceptions of crowding in protected areas by analyzing two new variables: *driving distance* to the trailhead and the *quality of the viewpoint*. The variable on *driving distance* provides insight on the likelihood that visitors may displace to alternative sites, and by measuring the *quality of the viewpoint* we can learn how visitors may be willing to trade other elements of their outdoor experience for the reward of a quality view. In short, we aim to improve our understanding of how visitors make tradeoffs across multiple dimensions, with particular attention to crowding, in order to assist park managers with crowding policy and management.

2. Literature review

Visitor crowding has a long history in studies on outdoor recreation (Manning 2003; Stewart and Cole 2001; Wagar 1964). Researchers have sought to understand the perception, impacts and consequences of crowding at national parks and recreation areas (Manning *et al.* 2002). Crowding has also received attention in the tourism and visitor attraction literature (Leask 2016). Crowding can reduce visitor satisfaction and become a problem for park managers who are interested in maintaining a high quality experience for all visitors.

To address crowding problems in parks and recreation areas, scholars have proposed developing standards of quality, or carrying capacities, for each recreation area (Manning 2001, 2003). However, the notion of carrying capacity and user limits remains contested, as some authors argue that the relationship between the number of encounters and visitor satisfaction is weak (Stewart and Cole 2001). The methodological challenges to managing perceptions of crowding make this debate particularly difficult to resolve (Manning *et al.* 2002).

The literature identifies at least three reasons why it is difficult to study the link between crowding and visitor satisfaction. First, measuring perceptions of crowding is difficult because visitors often employ mental and attitudinal coping mechanisms that can change perceptions over time or across circumstances (Arnberger and Haider 2007; Hall and Shelby 2000; Hall and Cole 2007; Hoss and Brunson 2000; Johnson and Dawson 2004). Coping mechanisms can include (1) displacement, in which visitors respond to crowding by using different areas of the park or changing their temporal patterns of park use (Arnberger and Haider 2007; De Freitas, Sutton, and Moloney 2013; Hall and Shelby 2000; Hoss and Brunson 2000); (2) “rationalizing,” in which visitors update or rationalize their opinions of an experience in a way that it becomes positive, despite potentially negative elements of crowding (Hall and Cole 2007; Hoss and Brunson 2000); and (3) employing “product shift,” changing their understanding of what the experience ought to be in order to accept the current situation (Hoss and Brunson 2000).

The adoption of these coping mechanisms has been well documented in the literature. For example, in a study of Acadia National Park users, over 90% of those surveyed employed various coping mechanisms in response to increased usage in the park, such as displacement, rationalizing or product shift (Manning and Valliere 2001). These authors emphasize that such coping mechanisms make crowding exceedingly difficult to identify in park satisfaction surveys.

A second reason the link between crowding and visitor satisfaction is difficult to study is that perceptions of crowding are influenced by multiple factors, not just by increased visitor density (Arnberger and Haider 2005; Arnberger and Brandenburg 2007; Arnberger and Haider 2007). In a study of an urban park in Austria, crowd-averse individuals reacted negatively to high and mixed uses of trails, while a crowd-tolerant group disliked *both* low and high use trails and instead preferred some mix of human encounters (Arnberger and Haider 2005). The study gave evidence that users experience crowding in different ways.

In a follow-up study at the same location, Arnberger and Haider (2007) measured how crowding caused displacement among visitors from an urban park. They found that group size, face-to-face encounters, presence of unleashed dogs, types of users and different speeds of users all influenced individuals' choices to displace. Interestingly, in some cases, *small* groups, specifically lone male users, caused displacement. This could be attributed to safety concerns where lone male users may elicit feelings of vulnerability (Arnberger and Haider 2007).

Scholars have also found that individuals perceive crowding differently according to their familiarity with an area (Arnberger and Brandenburg 2007). At the Danube Floodplains National Park in Austria, local visitors, or those more familiar with the park, were more likely to perceive the park as crowded and to displace than were visitors from the wider region, country or abroad (Arnberger and Brandenburg 2007). Cultural background, user motivations, expectations, user clustering and group size, and distance between encounters have also been shown to affect perceptions of crowding (Dawson and Watson 2000; Sayan *et al.* 2013).

Finally, a third reason the link between crowding and visitor satisfaction is obscured is that crowding is not the only determinant of a satisfactory outdoor experience. Even while researchers have developed methods to measure perceptions of crowding that account for coping mechanisms and the multiplicity of variables that affect perceptions of crowding (Manning *et al.* 1996; Sayan *et al.* 2013), critics point out that understanding crowding in isolation from other park attributes, such as ecological degradation or day use fees for example, does little to inform park management decisions (Lawson and Manning 2002).

This is because the social dimension – or the level of crowding – of a hiking experience is neither the sole determinant, nor always the most important determinant, of a satisfactory hiking experience. Some visitors may care more about quality views and the condition of the trail. Others may place more value on having freedom from rules and regulations. In short, studying and evaluating crowding thresholds alone does not acknowledge the interdependency of what Robert Manning calls the three dimensions of the protected area experience: the social dimension, the resource or ecological dimension, and the managerial dimension, all three of which help to define the visitor experience (Manning 1999).

To understand how crowding impacts visitor satisfaction, protected area management researchers have turned to discrete choice experiments (Arnberger and Haider 2007; Cahill, Marion, and Lawson 2008; Louviere and Timmermans 1990; Manning and Krymkowski 2010; van Riper *et al.* 2011). A discrete choice experiment is a stated preference method that uses surveys to ask respondents to “make choices among alternative configurations of a multi-attribute good” (Lawson and Manning 2003, 356). This allows researchers to model how the public will respond to hypothetical social, resource, and managerial conditions within a park. These alternative configurations, often referred to as scenarios, are populated with a series of attributes, typically reflecting diverse conditions within the park under study. As respondents make trade-offs between attributes and choose between the paired scenarios, they reveal their utility for each of the attributes.

Discrete choice experiments have two major advantages over other revealed preference methods. The first advantage centers on the use of hypothetical scenarios. Exposing respondents to hypothetical scenarios through a discrete choice experiment allows the researcher to explore the respondents’ perceptions of potential future situations (Haener, Boxall, and Adamowicz 2001). This ensures that all participants are exposed to the same scenarios and alternatives, and thus allows for a controlled analysis of how a respondent’s background (e.g. cultural, recreational, level of familiarity with an area, etc.) influences how they value the attributes presented. In addition, the use of hypothetical scenarios avoids triggering the coping mechanisms that respondents use to rationalize real world experiences.

The second major advantage of discrete choice experiments is that they allow researchers to isolate how particular attributes of a good contribute to an individual’s valuation of that good. As a result, choice experiments are effective tools for studying how ecological, social, and managerial attributes define the outdoor recreation experience. As Hanley and colleagues suggest, “choice experiments seem to be ideally suited to inform the choice and design of multidimensional policies” (Hanley, Mourato, and Wright 2001, 452).

Lawson and Manning (2003) used a stated choice survey to predict public support for restricting visitor use in the remote, backcountry area of Denali National Park. The study asked respondents about tradeoffs they would be willing to make between a solitary wilderness camping experience and stricter limits to access through the use of a permit system. They found that 75% of respondents would prefer limited access in order to reduce levels of crowding within the park (Lawson and Manning 2003).

Cahill, Marion, and Lawson (2008) also used a stated choice study to examine visitor tradeoffs in fairly accessible front-country areas versus more remote, backcountry areas of Acadia National Park. The results differed based on the location. In front-country areas, respondents preferred a managerial approach that would increase recreational “development,” the hardening of trails and the construction of boardwalks to

accommodate increased visitor use. However, in backcountry areas, respondents preferred limiting access and limiting development of recreational facilities (Cahill, Marion, and Lawson 2008).

Many discrete choice experiments studying crowding in protected areas have adopted the use of visual images in their surveys (Cahill, Marion, and Lawson 2008; Manning *et al.* 1996; van Riper *et al.* 2011). The use of digitally altered images of park sites featuring a range of different levels of visitor use allows researchers to present respondents with more realistic portrayals of what crowding might look like and avoid awkward textual language describing the scenarios (Manning *et al.* 1996; van Riper *et al.* 2011). The previously mentioned study by Cahill, Marion, and Lawson (2008), for example, used images to communicate what “development” of natural areas and trail hardening might look like to visitors (2008). Manning *et al.* (1996) found that visitors identified higher tolerances for crowding when presented with representative images of a scenario than they did if merely asked how many other visitors they would tolerate at the same site. Van Riper *et al.* (2011) used images to represent managerial approaches for regulating high use levels (e.g. fencing) as well as ecological degradation on and off trails. Pettebone *et al.* (2011) used an image-based stated choice experiment to model visitor preferences for transportation mode in the Rocky Mountain National Park, finding that crowding is a sufficient annoyance that park users are willing to change travel mode choice, from private vehicles to public shuttles, to avoid crowds and minimize visitor congestion.

Researchers have further built on discrete choice experiments by incorporating latent class analysis, which can be used to understand the heterogeneity of visitor preferences for crowding levels and other attributes. Latent class models are used to show these heterogeneous preferences among different sample population subgroups – or latent classes. Latent class modeling has been used to understand many issues in outdoor recreation. Hunt, Haider, and Botton (2005) studied the varying preferences of moose hunters in Northwestern Ontario, while Beardmore *et al.* (2013) used latent classes to predict the behavior among subgroups of fishers in Mecklenburg-Vorpommern, Germany. The use of latent class analysis to understand perceptions of crowding in protected areas is rarer compared to its use in other outdoor recreation studies. Juutinen *et al.* (2011) used latent class analysis to model heterogeneous visitor preferences for biodiversity levels, visitor numbers, and park development for a protected area in Finland (Juutinen *et al.* 2011). Wallmo and Edwards (2008) evaluated latent classes to model the value of species and habitat protection in potential marine protected areas.

In sum, an increase in the recreational use of a park can raise difficult questions for park managers. Is the increase in use leading visitors to feel crowded and reducing visitor satisfaction? Are visitors displacing to other sites and would more displace if limits to access were imposed? Might some users appreciate policies that restricted visitor use, such as user fees? And finally, are all visitors impacted by crowding in the same way or are some disproportionately more impacted? By answering these questions, we aim to contribute to the broader discussion about crowding management in parks and recreation areas.

3. Study area

Garibaldi Provincial Park (hereinafter referred to simply as Garibaldi Park) is situated in the rugged Coast Mountain Range of British Columbia (BC). The Park covers 194,000 hectares and is located in the Sea to Sky Corridor, 75 kilometers north of the Vancouver

metropolitan region, which contains over 2 million residents (Statistics Canada 2012). Garibaldi Park is a favorite outdoor destination for hikers, backpackers, backcountry skiers, mountain bikers, heli-skiers, climbers, and anglers and is considered one of the most popular parks offering backcountry hiking and camping experiences in the Vancouver area (CBC News 2016). BC Parks has a total of 30 provincial parks offering backcountry camping experiences. Only three, including Garibaldi, require online reservations for backcountry camping in order to manage the number of people using the sites (BC Parks 2016).

The number of people using Garibaldi Park and other BC Parks has become a concern for park managers and visitors. Between 2011 and 2013, visitor use of BC Parks increased by 7.6% (BC Parks 2013a). During this same period, the South Coast, which includes Garibaldi Park, saw a jump from 6 million to 7 million day use visitors (BC Parks 2013b). Garibaldi Park observed a 21.1% increase in camping visits, though this increase may be attributed partially to improved reporting methods (BC Parks 2013a). On a typical summer weekend day, Garibaldi Park's most popular trailhead can host up to 500 day use hikers and trailhead parking lots are often overflowing with vehicles (Baluja 2016; Kohlhardt 2015).

Visitors to Garibaldi Park have expressed frustration with overcrowding in online forums, as well as in personal comments to BC Park staff (Spearhead Huts Society 2012; BC Park Staff, personal communication, 24 March 2014). One visitor described Elfin Lakes, a popular overnighting area located in the Diamond Head region of Garibaldi Park, as a "circus scene on the weekends" (Spearhead Huts Society 2012). Park rangers regularly report instances of crowding and the "hardening" – compaction of vegetation and soils into hard, bare earth – of camping areas (BC Park Staff, personal communication, March 24, 2014). Indeed, based on this anecdotal evidence, BC Park officials estimate that use in some areas within Garibaldi Park has increased by up to 300% over what they consider to be the capacity of these areas (BC Park Staff, personal communication, 14 February 2014). A recent report indicates that park usage in our study area might be driven by social media, as an increasing number of young users seek to visit dramatic landscapes in their quest to capture a provocative selfie, a stunning profile shot, or viral image that would be widely shared among peers (Baluja 2016).

We examined visitor preferences for levels of crowding at the three most popular recreation areas within Garibaldi Park: the Elfin Lakes/Diamond Head area, the Black Tusk / Garibaldi Lake area, and the Wedgemount Lake area. Each site will be referred to hereafter as *Diamond Head*, *Garibaldi Lake*, and *Wedgemount* (Figure 1). Diamond Head, Garibaldi Lake, and Wedgemount are three of the five trailhead access points to Garibaldi Park. Park staff have identified both the Diamond Head and Garibaldi Lake areas as experiencing the most challenges with crowding.

4. Methods

4.1. Choice experiments theoretical background

Discrete choice experiments have their theoretical basis in psychology, economics, and statistics (Hoyos 2010), such as Lancaster's consumer theory (Lancaster 1966) and random utility theory (RUT) (McFadden 1974; Train 2003). RUT assumes that when a respondent or consumer chooses from different alternatives he/she evaluates these comparatively and chooses the option that maximizes their personal utility (Louviere, Hensher, and Swait 2000).

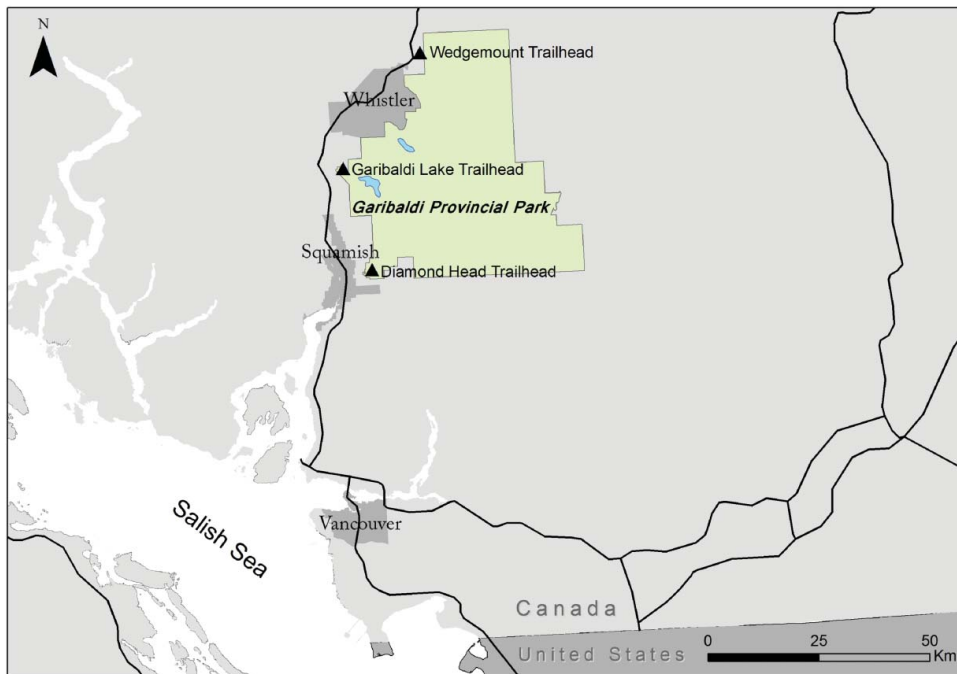


Figure 1. Location of Garibaldi Provincial Park in British Columbia, Canada. Source: the authors.

A choice experiment is implemented through a survey to a population of interest, in which respondents make choices from a series of alternative scenarios. The survey responses can be analyzed with a conditional logit model, which is able to observe the effect that the scenario attributes have on respondent choices (Boxall and Adamowicz 2002). The conditional logit model assumes homogeneity in preferences across respondents, and, therefore, limitations exist surrounding the explanatory power of the model (Colombo, Hanley, and Louviere 2009). As a result, segmentation approaches, such as latent or known classes, have been developed in order to account for this heterogeneity. In this study we used a latent class approach.

4.2. Latent class segmentation

Consumer or public preferences for goods and services are characterized by heterogeneity. Incorporating and understanding heterogeneity provides information on the potential impact of resource use decisions or policy impacts (Boxall and Adamowicz 2002). One approach is the use of random parameter logit/probit models which explicitly account for heterogeneity by allowing model parameters to vary randomly over individuals (Train 1998). However, these methods are not well-suited to explaining the sources of heterogeneity based on the characteristics of the individuals (Boxall and Adamowicz 2002).

An alternative to this approach is the use of latent class analysis, originally introduced by Lazarsfeld (Lazarsfeld 1950). Latent class analysis couples information on preferences from the choice experiment and observed measures (e.g. attitudinal measures, income, etc.), offering an opportunity to understand preference heterogeneity in consumer demand analysis (Boxall and Adamowicz 2002). The resulting latent classes

are unobservable subgroups or segments, in which the cases within the same latent class are relatively homogeneous on certain criteria, while cases in different latent classes are dissimilar from each other (Vermunt and Magidson 2005). Therefore, in the case of discrete choice experiments, the latent classes reflect differences in the parameters across unobserved subgroups for one or more attributes.

4.3. Discrete choice experiment online survey

We used a discrete choice experiment survey to evaluate visitor preferences and tradeoffs between crowding levels, managerial approaches, and natural resource conditions. We developed 96 hypothetical scenarios that were depicted through a combination of visual images and text. We created the visual images in Adobe Photoshop with real photographs from the region. The depicted scenarios were not meant to *specifically* represent the Diamond Head, Garibaldi Lake, or Wedgemount areas. Rather, the pictures were representative of terrain in BC to present realistic scenarios for respondents.

Based on conversations with BC Parks staff and on a review of other park studies (Cahill *et al.* 2008; Lawson and Manning 2002; Van Riper *et al.* 2011), we developed outdoor recreational experiences with variation along six attributes: viewpoint crowding; trail crowding; viewpoint conditions; trail conditions; day use fee; and driving distance to trail (Table 1). Each attribute had four levels, except for viewpoint conditions which had two, either “destination worthy” or “non-worthy.” We also altered the background imagery of the scenarios, since we expected that background scenery would influence visitor preference. We selected viewpoint crowding, trail crowding, trail conditions, and day use fees to represent the social (crowding), resource or ecological (trail conditions), and managerial (day use fees) dimensions of the outdoor recreation experience. Other studies have also selected attributes based on their representativeness of these three dimensions in order to model how visitors make tradeoffs between the three dimensions (Cahill *et al.* 2008; Lawson and Manning 2002; Van Riper *et al.* 2011)

We asked respondents to imagine they were searching for hiking options for an upcoming weekend outing, and we provided them with pictures and information regarding two hypothetical hikes in the Sea to Sky Corridor of BC. We asked respondents to select the recreational experience that most appealed to them.

We designed the discrete choice experiment to evaluate crowding, both on trails and at viewpoints, because we expected visitors to have different crowding tolerance levels at both locations. As previous studies have shown (Arnberger and Haider 2005; Arnberger and Haider 2007), individuals can feel more or less crowded depending on how they encounter other parties: whether they are moving fast or slowly, whether they are encountered face to face, or how large the other party is. Furthermore, we expected certain groups of visitors would tolerate crowded trail conditions in order to access particularly worthy viewpoints, while other groups – those more dedicated to the solitary wilderness experience – would be less tolerant of crowded trail conditions despite the viewpoint rewards.

We developed the four levels of crowding on the trails and at viewpoints to show the range of park visitors who could be visible at any one point. BC Parks Staff confirmed that the range of crowding levels selected for this study were realistic (Kohlhardt 2015). This range is quite wide as the Park can be both quiet during rainy weekdays and extremely busy on sunny weekend days.

We based the four levels of the *trail conditions* attribute on real trails in Garibaldi Park and within the region. While Garibaldi Park has some natural-looking single-track

Table 1. Attributes and levels of the discrete choice experiment for Garibaldi Park users.

Attribute	Level	Attribute	Level
Viewpoint crowding (communicated through images)	0 people per 100 meters	Day use fee	No access fee
	6 people per 100 meters		\$5
	12 people per 100 meters		\$10
	24 people per 100 meters		\$18
Trail conditions (communicated through images)	Natural appearance and little erosion	Driving distance to trail	70 km
	Moderate erosion, including some trail braiding		100 km
	Managed trail characterized by steps and/or stone and log placement intended to prevent erosion and trail braiding.		140 km
	Extreme erosion		180 km
Trail crowding (communicated through images)	0 people per 100 meters	Background scene on trail	Forested trail small foliage
	6 people per 100 meters		Forested trail large trees
	12 people per 100 meters		Alpine trail mountain
	24 people per 100 meters		Alpine trail meadow
Viewpoint conditions (communicated through images)	Worthy destination (i.e., attractive views)	Background scene at viewpoint	Rock/mountain viewpoint
	Non-worthy destination (i.e., less attractive views)		Treed viewpoint

trails, its more popular trails tend to either be actively managed by BC Parks, marked by widening and smoothing, additions of logs to border the trails to guard the trails against erosion, or they tend to be moderately and even extremely eroded.

We selected the attribute levels for the *day use fee* to represent the current situation – where no *day use fee* is charged – as well as the upper limits of what BC Parks might consider charging. Ontario Parks, for example, charges up to \$17.70 CDN for a daily vehicle permit. We thus used \$18 CDN as the upper limit for the *day use fee* (Ontario Parks 2015).

We added *viewpoint conditions* as an attribute (i.e., whether or not the viewpoint was “destination worthy” or “non-worthy”) because we expected visitors’ outdoor recreational experiences to be significantly impacted by the quality of the view. For especially view-worthy destinations, we expected visitors to tolerate sub-optimal conditions of other attributes such as crowding levels or ecological integrity. In the context of Garibaldi Park, we expected viewpoint conditions to help inform tolerance levels of crowding at different areas within the park.

Finally, we selected *driving distance* as an attribute to represent the willingness of Park visitors to displace to other locations when confronted with conditions not to their liking. For Garibaldi Park specifically, we designed the discrete choice experiment to demonstrate the likelihood that visitors would displace to other provincial parks in the

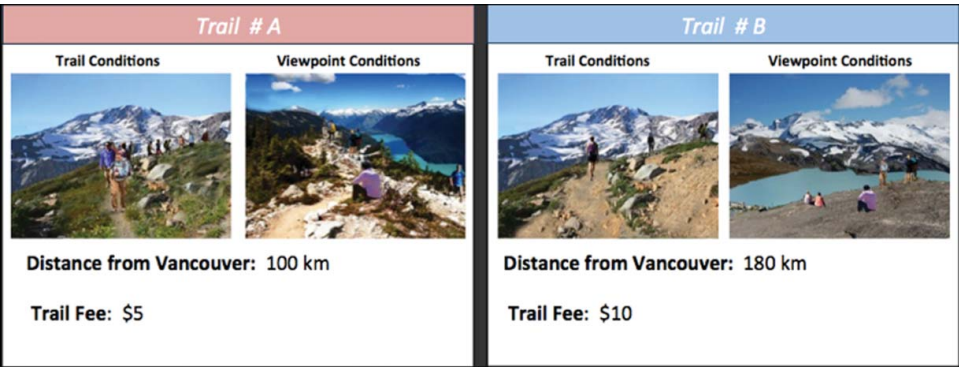


Figure 2. Example choice set used in the choice experiment. Garibaldi Park visitors were asked: “Which trail would you choose?”. Source: the authors.

vicinity of the Park. We selected the *driving distance* attribute levels because they represent real driving distances from Vancouver, BC, to three common trailheads within Garibaldi Park and to a fourth trailhead 180 km from Vancouver in Joffe Lakes Provincial Park.

A full factorial design of these six attributes (eight including the background scenes), with every possible pairing of attributes, would result in a total of 1,300 scenarios. To reduce the number of scenarios in the experiment to a manageable level, we used an orthogonal fractional factorial design to select 96 scenarios and 48 paired scenario profiles. All 48 profiles were then blocked into eight surveys, each with six of the profiles. An example choice set found in the discrete choice experiment survey is included in [Figure 2](#).

The discrete choice experiment survey also asked basic questions about visitors’ familiarity with the Park, demographics, level of expertise as outdoor recreationalists, and motivations for visiting the Park. To gauge their level of outdoor experience, we asked respondents to self-classify themselves as *casual*, *advanced*, or *committed* outdoor recreationalists. Similar self-classification survey questions have been used in other studies (Kerins *et al.* 2007). Motivations driving park visits are useful in outdoor recreation studies to identify different user groups and thereby assist in designing different management policies catering to these user groups (Chen and Jim 2012; Cheung and Jim 2013). In our case, the motivations helped to identify latent classes of park visitors, for the improved analysis of choice experiment data (Landauer, Pröbstl, and Haider 2012). A variety of different motivations were assessed, ranging from nature-based motivations to social-based motivations. The motivations used are adapted from a list used previously by Parks Canada in collaboration with Simon Fraser University’s Resource and Environmental Management program in their Chilkoot Trail National Historic Site Summer Recreation Use Survey (Légaré and Haider 2005).

We developed the discrete choice experiment survey using Fluid Surveys, an online survey software program that hosted our survey online. We recruited 10 personal acquaintances familiar with hikes in the Garibaldi Park region to trial test initial versions of the survey, and help us to assess respondent burden. We obtained ethics approval from the Behavioural Research and Ethics Board of the University of British Columbia.

We recruited survey participants at the previously mentioned Garibaldi Park trailheads – Diamond Head, Garibaldi Lake, and Wedgemount – from 1 August 2014 to 20 September 2014 (Kohlhardt 2015). We intercepted participants as they completed

their hike and asked them to participate in a short survey. Participant selection was based simply on the next Park visitor to exit the trailhead and enter the parking lot. During hours when the Park was quiet, this meant that we intercepted every visitor or group of visitors exiting the trailhead. During busy hours, we intercepted the next visitor upon completion of a previous survey. When we intercepted larger groups, we selected an individual respondent based on who in the group had the most recent birthday. We only recruited adults to participate in the intercept survey.

Upon completion of the intercept survey, we asked participants if they were willing to participate in an online follow-up survey. Of the 665 individuals intercepted, 527 gave their email addresses in order to participate in the online discrete choice experiment survey. We emailed these participants an invitation to participate in the discrete choice experiment survey at the beginning of October 2014 and asked them to complete the survey by December 2014. We sent three subsequent emails to remind participants to complete the survey. A total of 270 participants responded, providing us with a 51% response rate.

We analyzed discrete choice experiment responses with LatentGOLD Choice 4.0 software, which produces an extended conditional logit model for estimating the probability of respondents making specific choices among a set of alternatives as a function of the different attributes and attribute levels incorporated into the alternatives (Vermunt and Magidson 2005). LatentGOLD is commonly used as a choice modeling software by choice experiment experts (Landauer, Pröbstl, and Haider 2012). The conditional logit model reveals the general preferences of survey respondents. However, as discussed previously, conditional logit models do not account for the heterogeneity of visitor preferences. Consequently, we used a latent class model to achieve a more nuanced understanding of how different subgroups value the attributes under study.

4.4. Data refinement

We eliminated from the model responses from respondents that we suspected of not understanding the survey task, not taking the survey questions seriously, or of being significant outliers. Inclusion of outliers prevented us from achieving three distinct classes through the latent class analysis. These outlier respondents were identified through a two-step process, including first a principal component analysis with a varimax rotation of the “motivation” questions and second a cluster analysis of the principle component analysis. A total of 14 of the 270 respondents (5%) were eliminated as outliers from the cluster analysis.

We then linearized the numeric attributes within the model. The *distance*, *fee*, *trail crowding*, and *view crowding* attributes all consisted of four numeric, ordinal attribute levels. These levels, however, did not progress linearly from their lowest number to the highest. Fee, for example, had a low attribute level of \$0, increased to \$5, \$10, and finally, to \$18. Linearization of the attributes has been used in previous studies in order to achieve more interpretable results and to linearly graph respondents’ utility values with the resulting coefficient as the slope of the line (Landauer, Pröbstl, and Haider 2012). Nominal attributes were left as they were.

5. Results

We first describe the characteristics of our respondents; we then present the results of the conditional logit model for our entire sample; and finally, we then present the latent class analysis results, which allow us to look at different user types.

5.1. User profile

Nearly two-thirds (63%) of our respondents were male, with the remainder being female. Nearly half (47%) of the respondents were between 20 and 40 years old. The majority of respondents self-categorized their ethnicity as white (85%), followed by Chinese (5.3%), Filipino (1.9%), and Latin American (1.5%). The majority of respondents (87%) were from Canada and 8% were from the United States. Only 5% were from countries outside of North America. 62% of respondents were from the Metro Vancouver area, and 9% were “locals,” living within 40 kilometers of the Park. In terms of their level of outdoor experience, more than half (55%) of the respondents categorized themselves in the mid-range, *active* outdoor recreationalists, while 29% categorized themselves as *committed* recreationalists, and 15% reported themselves as *casual* recreationalists.

With regard to motivations for visiting the Park, *scenic beauty* was the most important motivation for the majority of respondents. *Enjoying the sights and smells of nature* and *experiencing the tranquility and peace* of the Park also ranked highly as primary motivations for respondents. The least important motivations were *spending time with family* and *being unconfined by rules and regulations*.

5.2. Conditional logit model

The results of the conditional logit model are shown in Table 2. The coefficients represent the regression estimates, referred to as part-worth utility parameters, that the model obtained for every attribute and attribute level.¹ The size of each coefficient shows the importance of each attribute for respondents. The coefficients for nominal attribute levels should be interpreted relative to each other as a result of being effects coded. Consequently, the magnitude of the difference between one level and another corresponds to the difference in part-worth utility between those levels. Therefore, the least preferred attribute level has the largest negative coefficient, and the most preferred attributed level has the largest positive coefficient.

Understanding visitor preferences for these different attributes and their levels reveals which characteristics matter most to Garibaldi Park visitors. The participants of this study were most concerned with *viewpoint crowding* (Tables 2 and 3), since this attribute shows the largest negative drop in utility (−1.93)². Three other attributes show a strong effect on respondents' preferences. *Trail crowding* (−1.30) is the second most relevant attribute, followed by the payment of a *day use fee* (−1.26), and then *viewpoint conditions* (worthy or unworthy) (0.80). In short, when evaluated as a single, homogenous group, respondents cared most about crowding in the Park and less about day use fees or worthy viewpoints.

We also found other variables with a smaller influence on a user's choice of trail. We find that visitors are less concerned about the *trail conditions* (0.61), as only the trails with a *natural appearance* and the *eroded trail* show any statistical significance. Moreover, the coefficients for the *trail background scenes* (0.55) obtained similar magnitudes to the previous attribute, and showed a preference for alpine trails rather than forested trails. We also find that respondents' preferences were less affected by the *driving distance* attribute (0.32). In other words, when given the choice of different hiking opportunities, Garibaldi Park visitors are willing to drive as much as needed in order to reach their ideal destination: fewer crowds, no day use fees, great alpine views, and little trail erosion.

Table 2. Estimates for crowding and management preferences (conditional logit model).

Attributes	Estimate	s.e.
Trail crowding, linearized	-0.432**	0.041
Viewpoint crowding, linearized	-0.643**	0.047
Day use fee, linearized	-0.420**	0.049
Trail conditions		
(1) Natural appearance and little erosion	0.331**	0.078
(2) Moderate erosion, including some trail braiding	-0.105	0.076
(3) Managed trail characterized by steps and/or stone and log placement intended to prevent erosion and trail braiding	0.053	0.075
(4) Extreme erosion	-0.279**	0.078
Viewpoint		
(1) Worthy destination (i.e., attractive views)	0.402**	0.048
(2) Non-worthy destination (i.e., less attractive views)	-0.402**	0.048
Driving distance to trail, linearized	-0.107**	0.042
Background scene attributes		
Background scene at viewpoint		
(1) Rock/mountain viewpoint	-0.015	0.048
(2) Treed viewpoint	0.015	0.048
Background scene on trail		
(1) Forested trail switchback	-0.318**	0.072
(2) Forested trail chief	-0.052	0.082
(3) Alpine trail mountain	0.234**	0.076
(4) Alpine trail meadow	0.136*	0.081
Constants		
1	0.078**	0.032
2	-0.078**	0.032

*Significant at the 10% level.

**Significant at the 5% level.

5.3. Latent class analysis

Our latent class analysis identified three latent classes of park users, each with distinct preferences for their hiking experience (Tables 3 and 4). The three user groups can be distinguished by their sensitivity to changes in viewpoints, their reaction to a hypothetical day fee, and the trail conditions. Given that other latent class models did not show a significant difference regarding the *driving distance* attribute between latent classes, and due to the fact that the estimates were relatively similar when compared to other attributes within each class, we restricted the model definition of “*driving distance*” in order to hold it constant across all three classes. This restriction still allows for the evaluation of the contribution of this *driving distance* attribute to the preferences within each of the classes.

The resulting classes of the respondents can be described based on three significant covariates related to outdoor experience and personal motivations. The hiking experience variable incorporates the self-classified outdoor recreation category that respondents reported with the frequency that each respondent went hiking ($p = 0.01$). The other two covariates describe their motivations to engage in hiking, either to perform an activity

Table 3. Estimates for all levels of numeric variables for all models.

Attributes and levels	Conditional logit model	Latent class 1	Latent class 2	Latent class 3
Trail crowding, linearized				
0	0.57	0.83	0.76	1.47
6	0.24	0.36	0.32	0.63
12	−0.08	−0.12	−0.11	−0.21
24	−0.73	−1.07	−0.97	−1.89
Viewpoint crowding, linearized				
0	0.84	1.52	0.46	3.51
6	0.36	0.65	0.20	1.50
12	−0.12	−0.22	−0.07	−0.50
24	−1.08	−1.96	−0.60	−4.51
Day use fee, linearized				
0	0.58	−0.38	1.37	2.37
5	0.23	−0.15	0.54	0.93
10	−0.12	0.08	−0.29	−0.50
18	−0.68	0.45	−1.61	−2.80
Driving distance to trail, linearized				
70	0.15	0.13	0.13	0.13
100	0.07	0.06	0.06	0.06
140	−0.05	−0.04	−0.04	−0.04
180	−0.17	−0.15	−0.15	−0.15

with the family ($p = 0.03$) or to learn about nature ($p = 0.01$). In the following section, we use the particular reaction of each class to hiking conditions (utility estimates of the attributes and levels) and their respective motivations (covariates) to explain the differences between classes. We also illustrate the general behavior of the members of the latent classes using descriptive labels.

Class 1: the outdoor tourist

Class 1 (36% of sample) represents a demographic that cares most strongly about the level of *crowding at viewpoints* (−3.48), and they are also very concerned about *crowding on the trail* (−1.90)₂. Whether or not the *viewpoint has a quality view* (1.61) was the third most important attribute for this class. *Driving distance* (−0.28) resulted in having the lowest estimate and influence in the preferences of this group. Regarding covariates, this group has an intermediate level of experience with outdoor recreation relative to the other two groups. Based on their motivations, this segment is more motivated to visit Garibaldi Park to *learn about nature*, and cares the least about visiting the Park in order to *spend time with family*.

Compared to the other two classes, Class 1 cares the most about the *quality of the viewpoint*, and they have medium sensitivity to crowding at that viewpoint compared to the other groups. Based on these characteristics and preferences, we interpret this class to be a group of Garibaldi Park visitors with a burgeoning level of exposure to outdoor adventures and who are motivated strongly by their desire to find stunning views and to

Table 4. Results of the three class latent class model.

Attributes	Class 1	s.e.	Class 2	s.e.	Class 3	s.e.
Trail crowding, linearized	-0.632**	0.128	-0.577**	0.106	-1.118**	0.251
Viewpoint crowding, linearized**	-1.159**	0.179	-0.353**	0.113	-2.675**	0.627
Day use fee, linearized**	0.277*	0.152	-0.993**	0.172	-1.721**	0.407
Trail conditions**						
(1) Natural appearance and little erosion	-0.018	0.207	-0.210	0.179	3.321**	0.884
(2) Moderate erosion, including some trail braiding	-0.380*	0.199	-0.0002	0.227	-0.807*	0.418
(3) Managed trail characterized by steps and/or stone and log placement intended to prevent erosion and trail braiding	0.046	0.176	0.346*	0.209	0.189	0.399
(4) Extreme erosion	0.353*	0.209	-0.135	0.242	-2.703**	0.769
Viewpoint conditions **						
(1) Worthy Destination (i.e., attractive views)	0.803**	0.128	0.408**	0.110	0.394*	0.208
(2) Non-worthy destination (i.e., less attractive views)	-0.803**	0.128	-0.408**	0.110	-0.394*	0.208
Driving distance to trail, linearized	-0.094*	0.057	-0.094*	0.057	-0.094*	0.057
Background scene attributes						
Background scene at viewpoint**						
(1) Rock/mountain viewpoint	-0.234*	0.124	0.458**	0.119	-1.217**	0.383
(2) Treed viewpoint	0.234*	0.124	-0.458**	0.119	1.217**	0.383
Background scene on trail**						
(1) Forested trail switchback	-0.057	0.180	-0.783**	0.200	-2.061**	0.745
(2) Forested trail chief	-0.127	0.198	-0.099	0.189	0.329	0.576
(3) Alpine trail mountain	0.442**	0.172	0.456**	0.187	0.237	0.353
(4) Alpine trail meadow	-0.258	0.208	0.427**	0.211	1.495**	0.473
Constants**						
1	0.151*	0.080	-0.039	0.074	0.468**	0.188
2	-0.151*	0.080	0.039	0.074	-0.468**	0.188
Class size	0.356			0.347		0.297
Model for classes						
Intercept	Class 1	s.e.	Class 2	s.e.	Class 3	s.e.
	-0.235	0.375	-0.158	0.391	0.394	0.346
Covariates						
Hike frequency/recreation category						
Casual	0.0764	0.1377	0.367**	0.152	-0.444**	0.159
Committed	-0.0764	0.1377	-0.367**	0.152	0.444**	0.159
Family motivation	-0.2037*	0.1212	-0.163	0.129	0.367**	0.142
Learn about nature motivation	0.3492**	0.1607	0.209	0.170	-0.558**	0.189

*Significant at the 10% level.

**Significant at the 5% level.

learn more about nature. They are committed to the value of these outdoor experiences, and yet they are not as familiar with the ecological processes of natural areas.

The latent class model shows a few surprising results for this group regarding *day use fees* and *trail conditions*. First, this segment expressed having a positive preference for paying a *day use fee* (0.83). This result suggests members of this group support paying for park services. It may also be that the parameters we used to express the fee attribute (\$0–\$18) did not reach the limits of this latent class's upper willingness to pay. Presumably, their positive preference for a fee would become negative if the fees were increased to a higher rate. The other unexpected result for this segment is that they expressed - a strong preference for *eroded trails* over *managed trails* or trails in more *natural conditions* (0.73). However, pictures of eroded trails also show wider trails, in such a way that this finding may be a result of this group simply preferring wider trails.

Class 2: the casual recreationalist

This latent class (35% of the sample) has a very strong aversion to *day use fees* (−2.98), and this dislike for fees was the strongest preference expressed by this group. Also, *trail crowding* (−1.73) is more important than *viewpoint crowding* (−1.06) for this group, and relative to the other two groups, Class 2 cares the least about *crowding at viewpoints*.

In addition, having worthy *views at viewpoints* (0.92) and along segments of the trail has less importance for this class, compared to the previous attributes. They also expressed a slight preference for *managed trails* (0.56) over *single-track* or *eroded trails*. When faced with other trade-offs, this latent class reacted least to *driving distance* (−0.28). Regarding experience and motivational covariates, this segment has the least experience with hiking or the outdoors, and *spending time with family* and *learning about nature* are not significant motivations for visiting Garibaldi Park.

We interpret this class to be a beginner group of outdoor recreationalists who have not yet formulated strong opinions about the conditions of their hiking experience relative to the other two classes. They are affected by paying a day use fee, and they are more concerned with their trail experience (in terms of crowding, trail condition, and trail landscape) than they are with viewpoints. In other words, they lack the 'tourist' commitment that the first group possesses for seeking out viewpoints regardless of fees and trail conditions.

Class 3: experienced, freedom-seeking outdoors-person

The third class (30% of sample) reacted the most to *crowding at viewpoints* (−8.03). The group's second strongest preference was for *natural, single-track trails* (6.02), and they had a strong aversion to eroded trails. Also, they rejected the idea of paying *day use fees* for their hiking experience (−5.16). Ultimately, these results portray a group of park users that expect hiking opportunities that offer a natural experience away from crowds, without the need for fees.

Compared to other attributes examined, *crowding on the trail* (−3.35) had the smallest effect on this group's choices, and this class was not concerned as much with the condition or *quality of viewpoints* (0.79), although they strongly prefer trails through alpine landscapes. Similar to the other two classes, *driving distance* had the lowest influence (−0.28). Based on the covariates of this class, this segment is the most experienced of the three classes. With regard to motivations, this group is not concerned

about *learning about nature*, but they do visit the Park to *spend time with family* more than the other two groups.

We interpret this group to be traditional outdoor recreationalists who value the solitude and the freedom from fees or regulation (e.g. managed trails) in their outdoor experience. They care little about learning more about nature because they likely already possess knowledge of natural processes. Presumably, when these respondents hike with family, they choose accessible trails even if they are crowded. However, if fees are charged, they might be displaced to other recreation options. They desire a wilderness experience, one that shows fewer signs of other visitors, and they appreciate their entire journey in the out-of-doors rather than just valuing the viewpoints, which differentiates them from other classes.

6. Discussion and management implications

6.1. Crowding

Visitor crowding has been one of the most heavily-studied issues in outdoor recreation (Manning and Valliere 2001, 410). Researchers have focused particular attention on whether and how crowding affects visitor satisfaction. Some researchers argue that crowding has little influence on satisfaction (Stewart and Cole 2001), whereas others argue that crowding is problematic enough that standards of quality and “carrying capacities” should be developed for parks in order to limit visitor access (Manning 2001; 2003). In addition to questions of whether or not crowding matters in general, researchers have also examined whether crowding matters for some individuals more than others. One study identified crowd-averse and crowd tolerant individuals who react differently to different levels of crowding (Arnberger and Haider 2005), and another observed that **local visitors were more sensitive to crowding than were visitors from outside the region** (Arnberger and Brandenburg 2007). Other studies have shown that cultural background, user motivations, expectations, user clustering and group size, and distance between encounters can affect visitor response to crowding (Dawson and Watson 2000; Sayan *et al.* 2013).

Our findings support the argument that crowding has a relatively-large influence on visitor satisfaction, and also add nuance to our understanding of the complexity of the crowding problem. We found that visitors to Garibaldi Park were more sensitive to crowding than to any other feature of the outdoor experience that we measured. The results of our model show that visitors lose more utility from either trail crowding or viewpoint crowding than they do from other undesirable features of their experience, such as day use fees, eroded trail conditions, or non-worthy viewpoint destinations. Even the presence of desirable features, such as natural trail conditions with no erosion and worthy viewpoint destinations, are not by themselves enough to offset the decrease in utility that stems from crowding.

Our results also show that visitors are sensitive not only to the degree of crowding that they experience, but also to *where* they experience the crowding within the park. While visitors experience negative utility from both crowding along the trail and at the viewpoint, our model results indicate that visitors are even more sensitive to crowding at the viewpoint. The utility estimate for viewpoint crowding was roughly 50% larger than the estimate for trail crowding. This suggests that park managers should monitor viewpoint experiences even more carefully than trail experiences in order to maximize visitor satisfaction. For example, arranging destinations so that visitors have multiple

focal points that disperse crowds across sites for viewing, resting, photography and interpretive material might distribute people more evenly at the destination, and improve the overall experience.

These types of adjustments provide managers with potential solutions to crowding that can protect visitor satisfaction without necessarily having to restrict visitor access through the collection of day use fees or other restrictive mechanisms. Whereas Lawson and Manning (2003) found that three out of every four visitors to Denali National Park would support the use of measures to limit access as a means to reduce crowding, we found that day use fees had the third largest negative utility for Garibaldi Park visitors, behind only viewpoint and trail crowding. Day use fees were generally only supported by 36% of our respondents.

Those respondents were part of “The Outdoor Tourist” class of visitors that we identified through our latent class analysis. This part of our analysis adds additional nuance to our understanding of crowding and how the perceptions of crowding vary across different classes of park visitors. The Outdoor Tourist class cared more about viewpoint crowding and worthy destinations than they did any other attribute, which might help to explain why they appear to derive positive utility from day use fees. For this class of user, their satisfaction depends so heavily upon being able to enjoy an attractive view without having to fight crowds that they express support for (and an apparent willingness to pay) day use fees that would help to restrict access. The second and third classes of users (i.e., “The Casual Recreationalist” and the “Experienced, Freedom-Seeking Outdoors-person”) were the least and the most sensitive to crowding respectively, though neither class supported day use fees. The existence of different classes of users with different sets of preferences regarding crowding and how to manage it suggests that park managers need to carefully consider how actions to manage crowding affect these different groups of visitors.

6.2. Trail conditions

While we found that visitors are more sensitive to crowding than to other park attributes, park managers should not address crowding without also considering other park attributes at the same time (Lawson and Manning 2002). Other relevant attributes include trail conditions, viewpoints, driving distance to the trail, trail type, and landscape.

Some groups strongly prefer natural-looking trails while others prefer wider or more managed trails. Trail condition was the *most* important attribute for the Class 3 user. For park managers, this implies that managing or hardening trails to accommodate more visitors may displace those who prefer more narrow and unmanaged trails. This is an important consideration for park managers in other contexts who are developing park experiences to accommodate all types of users. It also supports certain park management frameworks that recommend ‘product differentiation’ such as the Recreation Opportunity Spectrum used commonly throughout North America (Newsome, Moore, and Dowling 2012).

6.3. Viewpoints

Similar to trail condition preferences, the way visitors value their viewpoint experience also supports experience differentiation to accommodate the heterogeneity of park visitors. While each class of user derived relatively high utility from attractive viewpoints, this was much more important to Class 1 users than to Class 2 or 3 users. With this in mind, park managers might choose to develop more secluded, single-track

trails with less attractive viewpoints to accommodate more experienced users and wide, managed trails with stunning viewpoints to attract less experienced users. This requires that park managers be both selective and creative in deciding future development projects within parks.

6.4. Driving distance

We found that park users were not as sensitive to driving distance in their selection of hiking experiences as they were to other attributes. Perhaps, once park visitors have committed to driving a minimum of 70 kilometers to reach their hiking destination, they care little if they have to continue to drive further to achieve their preferable destination. The insensitivity to driving distance suggests that visitors are likely to displace as soon as conditions within a given park prove not to their liking. For park managers, this can mean both that visitors might already be displacing now for less crowded options and that the imposition of increased day use fees could result in further displacement of park users.

6.5. Trail type and landscape

We found that park visitors preferred alpine trails with quality views over trails bordered by smaller foliage and smaller trees. They also preferred trails through mature forest over trails through post-disturbance forest, and alpine trails over forested trails in general. The implication for managers is that trail design should try to maximize alpine travel and minimize travel through disturbed landscapes to optimize the visitor experience. Our finding also strengthens the mandate for park managers for resource protection.

7. Conclusion

Managers of outdoor recreation areas are responsible for providing the public with accessible recreation opportunities while also maintaining an agreeable outdoor experience for everyone. However, large crowds in recreation areas can quickly deteriorate the quality of a recreation experience and reduce visitor satisfaction. In highly popular recreation areas, it may be difficult for park managers to ensure widespread accessibility and *simultaneously* guarantee a high quality experience. However, policies that restrict user access in order to reduce crowding may conflict with the mandate for public access. In the face of this dilemma park managers need information regarding how best to manage park access in order to maximize visitor satisfaction.

Understanding visitor preferences for the social, managerial, and natural resource attributes of their park experience helps to define the most appropriate indicators on which to base park management decisions. Our choice experiment shows that visitors to Garibaldi Park care most about having fewer people at park viewpoints. While trail experiences are also important, viewpoint conditions may have a greater influence on the overall satisfaction of Park visitors. Thus focusing on managing crowds at the destination may provide park managers with a strategy for maintaining overall satisfaction without limiting visitor access. To increase visitor satisfaction, park managers might manage viewpoints more carefully, or develop more destination sites, rather than developing entirely new trails and viewpoints elsewhere in the park.

Importantly, we show that park visitors have heterogeneous preferences for the attributes under study, and that certain management responses to crowding could result in disproportionate displacement of some visitors over others. For example, implementing

user fees would disproportionately displace more experienced outdoor enthusiasts. Thus, if park managers hope to offer outdoor recreational opportunities for all visitors, then they will have to carefully manage trail conditions and restrict crowding at viewpoints. Alternatively, park staff could explore differentiating their ‘product,’ or the types of hiking experiences they offer to suit the tastes of the three visitor groups we identified in this study. For example, park managers might offer a free hiking experience with a wider, pleasant trail for the Casual Recreationalist. For the Outdoor Tourist, they could charge for a hiking experience that would lead to high-quality viewpoints. Finally, for the Freedom-Seeking Outdoorsperson, they could offer a rugged, single-track hiking experience free of charge, but with less stunning views than for the Outdoor Tourist. In short, understanding the various preferences of visitor subgroups can help managers design different experiences in order to maximize overall visitor satisfaction.

Our findings also differ with respect to past research. Perhaps surprisingly, we do not find an association between crowd aversion and visitor familiarity to an area as others have (Arnberger and Brandenburg 2007). In other words, local users were not more crowding averse than visitors less familiar with the park. However, this may merely be an artifact of displacement already taking place in the park, in which crowd-averse locals have already displaced to more solitary trails. This also suggests a limitation in choice experiment surveys that take place only at the area of interest, since respondents have self-selected themselves to that location, and perhaps only crowd tolerant users remain on the popular trails.

We find that the surveyed users at Garibaldi Park are particularly sensitive to day use fees, and even small fees could result in reduced visitor use of over 50%. Furthermore, such a move might damage the public image of the park agency, and create political tensions. Our results suggest that while a park may be very popular, the implementation of user fees may not generate large revenues, since the public is likely to displace to free alternatives. Therefore park managers should not overestimate the potential revenues associated with implementing user fees.

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Disclosure statement

No potential conflict of interest was reported by the authors.

Notes

1. For continuous, numeric variables, data was effects coded and the resulting coefficient represents the slope of the linear effect. The nominal attributes were effects coded (Trail Conditions, Background Viewpoint, and Trail Scene Attributes), and a coefficient is shown for each attribute level.
2. In parenthesis we report the difference between levels with the highest and lowest part-worth utility estimates in order to show the maximum potential effect of each attribute on the preferences. For numerical variables we present in Table 3 the estimates for all levels after transforming back from the effects coded and linearized data.

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